

Routine Names

Quick Reference Guide

Rules

A routine is either a *function* or a *procedure*.

Functions

A function's primary purpose is to return a value. For example, `Math.sin(x)` returns the sine of x .

A function is named by the values it returns.

Special cases:

- Conversions: `[source]2[target]`
- Boolean value: The return value is phrased as a question
- Java getters (old style): `get[variable name]`

Procedures

A procedure's primary purpose is to perform an operation.

A procedure is named as: **verb object**.

The object can be omitted if it is clear by the context

Test Questions

Find an appropriate name for a routine given the following short descriptions. (In the following, "this" refers to the object to which the method belongs.)

- a. Calculate the hyperbolic tangent of its argument.

Answer: `tanh`

Explanation: A function is named by the value it returns.

- b. Print this invoice.

Answer: print

Explanation: A procedure is named with a verb denoting the action being performed. In this case, the object is omitted as it is clear from the context (this): invoice.print() is clearer and more direct than invoice.printInvoice().

- c. Convert a metric value (liters) into ounces.

Answer: litersToOunces or similar

Explanation: A conversion function is named as [source]2[target]

- d. Add a new free item to this invoice. It throws an exception if the item is not in the list of promotional or gift items.

Answer: addFree or addFreeItem

Explanation: A procedure is named with a verb and an object. In this case, the object cannot be omitted, because invoice.add(item, price) means to add some item presumably at a cost, which is different than invoice.addFreeItem(item). Special cases and error handling do not have to be included as a part of the name.

- e. Returns whether this document has already been formatted for printing.

Answer: isFormatted or isPrintFormatted

Explanation: A function is named from the value it returns, and a Boolean return value is expressed as a question.

- f. Add to this employee the W-2 form for the current fiscal year and return whether the employee was modified by this operation. The given W-2 form will be ignored in many circumstances, such as if the amount in box 2b is negative or if it does not refer to the current fiscal year. It throws an exception if the given W-2 form is invalid.

Answer: addCurrentW2

Explanation: although the routine returns a status code, its primary purpose is to perform the addition of the W-2 form. Therefore, it is

a procedure, and it is named as verb [object]. In this case, the object must be given because `employee.add(x)` is misleading compared to `employee.addCurrentW2(x)`. Special cases and error handling do not have to be included as a part of the name.

- g. Return the name of this employee.

Answer: name

Explanation: A function is named by the value it returns. In the old Java style, it would have been called `getName`, assuming that `name` is a private variable of `Employee`.

The rules cover most (but not all) cases. You can argue for a different routine name, but only after showing that the canonical names fail to do its job.